1 Find the value of $16^{-\frac{3}{2}}$

.....

(Total for Question 1 is 2 marks)

---

2 Make x the subject of the formula $3x + a = b(x + 5)$

….....….....

(Total for Question 2 is 3 marks)

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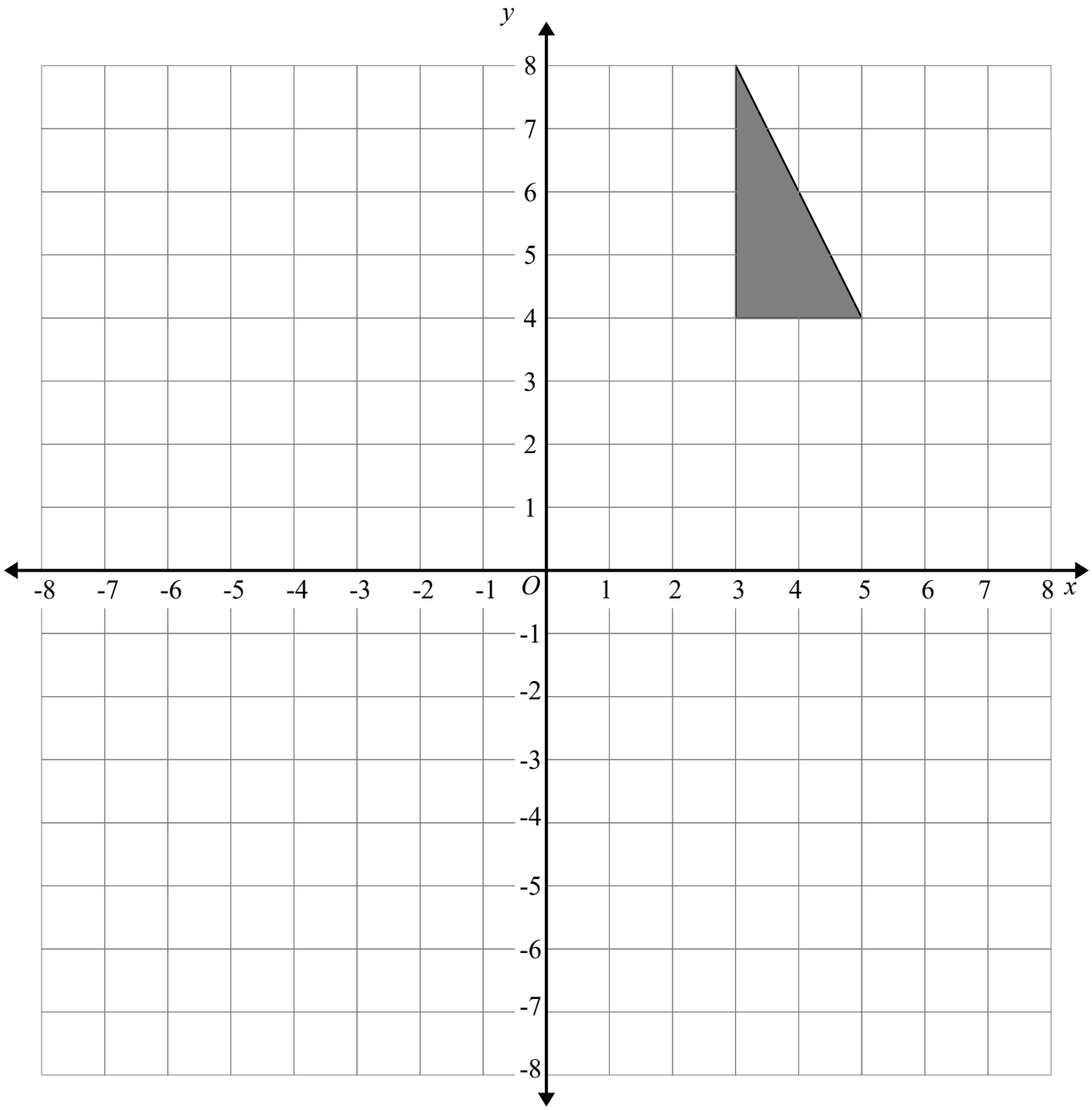
3 Expand and Simplify $(x + 5)(x - 2)^2$

….....….....

(Total for Question 3 is 3 marks)

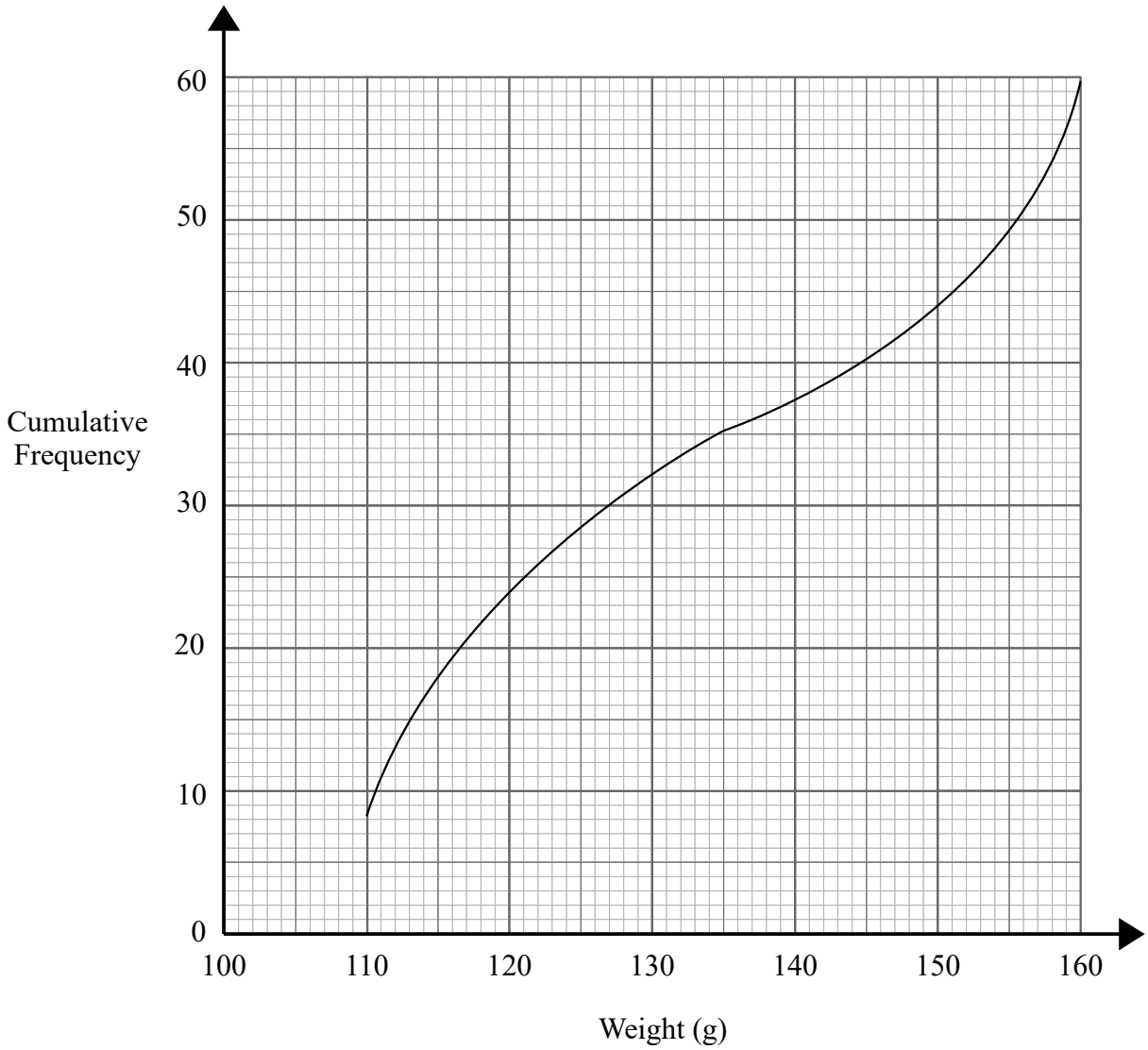
4



On the grid, enlarge the triangle by scale factor -1.5. centre (1, 2).

(Total for Question 4 is 2 marks)

5



(a) Find the median weight.

….....g
(1)

(b) Find the inter quartile range.

….....g
(2)

(Total for Question 5 is 3 marks)

6 Simplify fully $\dfrac{7 + 2\sqrt{5}}{2 + \sqrt{5}}$

….....

(Total for Question 6 is 3 marks)

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7 Prove algebraically that the recurring decimal $0.3\dot{1}\dot{8}$ can be written as $\dfrac{7}{22}$

(Total for Question 7 is 2 marks)

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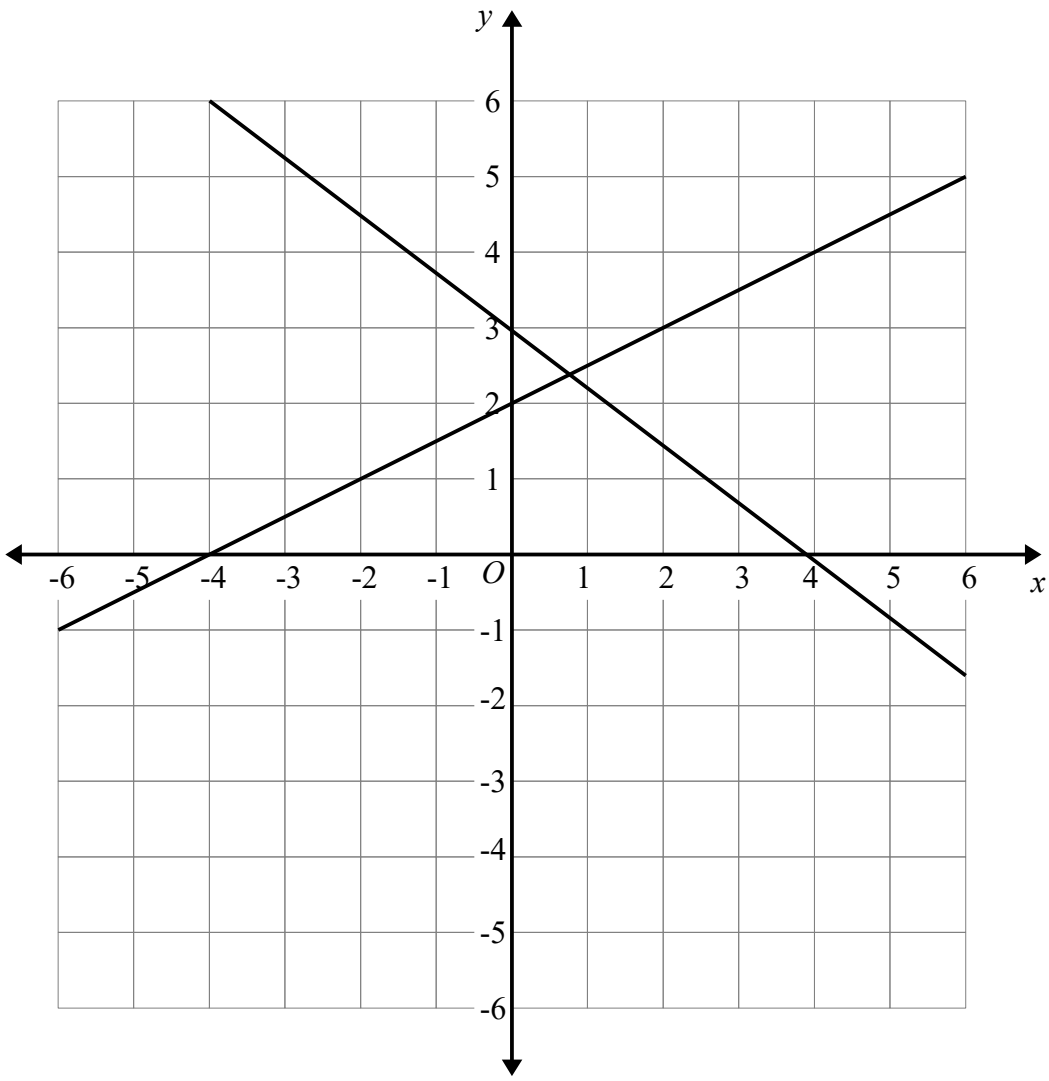
8 The graphs of the straight lines with equations $2y = x + 4$ and $4y + 3x = 12$ have been drawn on the grid.



x and y are both integers.

Mark with a cross (×) all of the points that satisfies all the inequalities

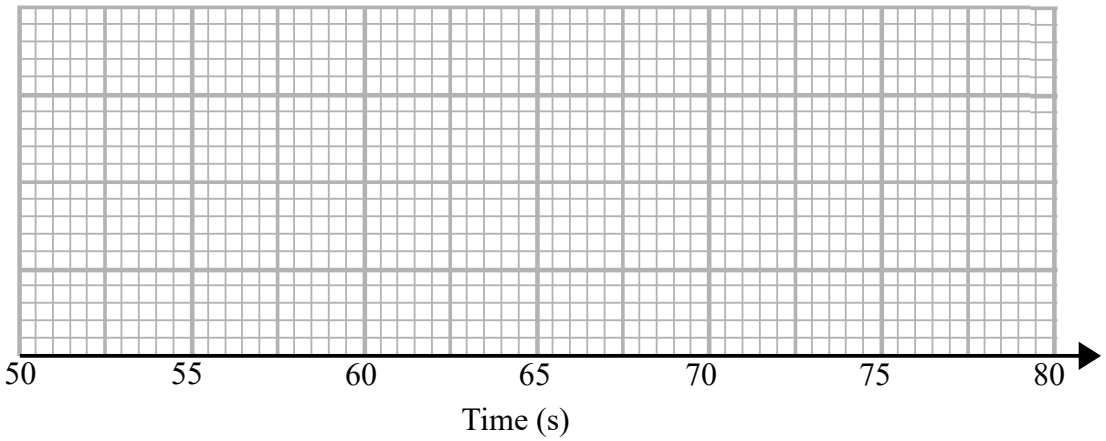
$2y > x + 4$ $\qquad$ $4y + 3x < 12$ $\qquad$ $x > -3$

(Total for Question 8 is 2 marks)

9 The times, in seconds, of 15 students running a race are recorded below.

53 54 57 58 58 60 61 65 67 67 68 70 71 71 78

Draw a box plot for this information.



(Total for Question 9 is 2 marks)

10 Line A passes through the points (2, -6) and (12, -1)
Find the equation of the line perpendicular to A that passes through (2,5)

….....….....

(Total for Question 10 is 2 marks)